

BSc

ARTIFICIAL INTELLIGENCE



Go a Step Further with Smart Tech

Artificial intelligence (AI) makes it possible for machines to learn from experience, adjust to new inputs and perform human-like tasks. Most AI examples that we hear about today – from chess-playing computers to self-driving cars – rely heavily on deep learning and natural language processing.

Using these technologies, computers can be trained to accomplish specific tasks by processing large amounts of data and recognizing patterns in the data in Health Care, Agriculture, Transportation, Engineering, Manufacturing, Retail, Banking, etc.

These advances in AI and Machine learning offer tremendous opportunities across the spectra of human endeavors and are poised to reshape future technologies and workforces.

Artificial Intelligence at ACity

In this program, we explore the ethical dimensions of AI, its strategic impact as well as the core programs and adjacencies such as robotics, informatics, and data analytics.

Our future industrialization efforts across the spectrum of key strategic areas such as manufacturing, healthcare etc. are pivoted around artificial intelligence, hence, we can no longer wait to join the AI and robotic wagon.

The aim of the BSc in Artificial Intelligence (AI) is to equip our graduates with knowledge in emerging advances in computational, decision-making sciences and technologies that allow computers and machines to function in an intelligent manner both in accurate prediction of events and outcomes.

Top Careers in Artificial Intelligence

Graduating students of this programme will be able to select, create, apply, integrate, and administer computing technologies in order to meet the needs of users within societal and organizational contexts. Some top career options include;

- 📌 Business Intelligence Developer
- 📌 Machine Learning Engineer
- 📌 Computer Vision Engineer
- 📌 Applications Developer
- 📌 Cyber Security Analyst
- 📌 AI Research Scientist
- 📌 Application Analyst

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for **Creative freedom
& Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics + Chemistry OR any other elective relevant to the chosen Programme)

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics + Chemistry OR any other elective relevant to the chosen Programme)

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and 3 passes in the A-Levels. (Elective/Add/Further Maths and Physics mandatory)

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects. (Elective/Add/Further Maths and Physics mandatory)

FRENCH IB

Minimum of 50% overall average pass. (subject to approval NAB) (Maths, English and Physics mandatory)

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths, English and Physics mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills
French Language
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I
Introduction to Programming with Python
Physical Sciences
Technology and Society
Pre-Calculus (with MATLAB)
Data Analysis using MS Excel

SEMESTER 2

Course Name

Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II
Logic and Critical Thinking
Text and Meaning
Analytic Geometry and Calculus I (with MATLAB)
Principles of Microeconomics
Programming in C
Financial Accounting I
Introduction to Artificial Intelligence and Robotics

SEMESTER 3

Course Name

Fundamentals of Innovation and Entrepreneurship (FIE) I
Leadership Seminar I
Analytic Geometry and Calculus I (with MATLAB)
Fundamentals of Logic Design
AI: Representation and Problem-Solving
Data Structures and Algorithms
Introduction to Computer Systems
Object Oriented Programming with C++

SEMESTER 4

Course Name

African Studies
Fundamentals of Innovation and Entrepreneurship (FIE) II
Probability, Statistics and Reliability (with MATLAB)
Applied Linear Algebra (with MATLAB)
Computer Architecture and Organisation
Design and Analysis of Algorithms
Signals and Systems

SEMESTER 5

Course Name

Leadership Seminar II

Applied Probability and Computing (with MATLAB)

Discrete Mathematics (with MATLAB)

Digital Signal Processing

Great Theoretical Ideas in Computer Science

Machine Learning

Object Oriented System Design

SEMESTER 6

Course Name

Project Management, Engineering Economics and Risk Analysis

Computational Neural Networks

Computer Vision

Industry Internship

Natural Language Processing

Research Methods in Computing

SEMESTER 7

Course Name

Autonomous Agents

Computational Perception

Intermediate Data Analytics

Introduction to Deep Learning

Project Phase I

Speech Processing

SEMESTER 8

Course Name

Professional Ethics and Values

Advanced Data Analytics

Intermediate Deep Learning

Design of Artificial Intelligence Products

Project Phase II